

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: DSA0204

- 1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.**

Ans:

Sampling is the process of converting a continuous image into a digital image by selecting pixels at regular intervals. The quality of the image depends on the sampling rate. If the sampling rate is too low, fewer pixels are captured, causing the image to appear blurred and lose important details. Fine edges and small objects may not be represented correctly. This reduces the overall clarity of the image. To correct this problem, a higher sampling rate or a higher-resolution camera should be used. Image interpolation and enhancement techniques can also improve the appearance of the blurred image.

- 2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.**

Ans:

Image sensing and acquisition are the first steps in creating a digital image. In low-light conditions, the camera sensor receives less light, making the image darker and increasing sensor noise. This reduces brightness, contrast, and visibility of details. Poor acquisition can also affect color accuracy and sharpness. The final digital image may therefore contain unwanted noise and blurred regions. Using a high-quality image sensor, increasing exposure time, adjusting camera settings, or adding external lighting can significantly improve image quality during acquisition.

- 3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.**

Ans:

Aliasing is a common image distortion that occurs when an image is sampled below the required sampling rate. As a result, edges appear jagged, and false patterns or unwanted shapes may be visible. Although quantization affects

intensity levels, aliasing is mainly caused by improper sampling. This problem can be reduced by increasing the sampling rate so that more image details are captured. Applying an anti-aliasing filter before sampling also removes high-frequency components that cause aliasing. These methods help produce smoother and more accurate digital images.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

Ans:

Pixel resolution refers to the number of pixels used to represent an image, while intensity resolution refers to the number of brightness levels available for each pixel. Higher pixel resolution produces sharper images and helps identify small structures such as blood vessels or tiny tumors. Higher intensity resolution allows the system to display slight differences in brightness, making it easier to distinguish between different tissues. In medical imaging, both types of resolution are equally important because they improve image quality and help doctors make accurate diagnoses.

5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Ans:

A surveillance system mainly works at the high level of Computer Vision because it performs intelligent analysis of captured images. The system first acquires images through cameras, then detects moving objects, identifies faces or vehicles, tracks their movement, and recognizes suspicious activities. Finally, it makes decisions such as generating alerts or recording events automatically. These tasks require understanding and interpreting visual information rather than simply processing images. Therefore, real-time surveillance is considered a high-level Computer Vision application used for security and public safety.

6. Explain how image formation models help in improving image quality in applications such as autonomous driving.

Ans:

Image formation models explain how light travels from an object to the camera sensor to create an image. These models help correct problems such as lens distortion, poor lighting, shadows, and perspective changes. In autonomous driving, clear and accurate images are essential for detecting lanes, traffic signs,

pedestrians, and other vehicles. By improving image quality, image formation models help the vehicle understand its surroundings more accurately. This increases the reliability and safety of self-driving systems.

7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Ans:

Quantization is the process of converting continuous grayscale intensity values into a fixed number of digital levels. Each pixel is assigned the nearest intensity value based on the available levels. If more quantization levels are used, the image appears smoother and preserves more details. If fewer levels are used, the image loses fine details and may show visible banding or false contours. Therefore, proper quantization is important for maintaining image quality and accurate representation.

8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Ans:

Noise is introduced during image acquisition due to low light, sensor limitations, electronic interference, or high camera sensitivity (ISO). Poor acquisition conditions reduce image clarity and make object detection difficult. Noise can be minimized by using better image sensors, improving lighting conditions, selecting proper camera settings, and reducing unnecessary electronic interference. Image processing techniques such as median filtering, Gaussian filtering, or noise reduction algorithms can further improve image quality after acquisition.

9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.

Ans:

Facial recognition uses different levels of Computer Vision to identify a person accurately. First, the camera captures the face image during image acquisition. The image is then preprocessed by removing noise and adjusting brightness or contrast. Important facial features such as the eyes, nose, and mouth are extracted and compared with stored data in a database. Finally, the system recognizes the person and makes a decision, such as granting access or confirming identity. Each stage plays an important role in improving recognition accuracy.

10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.

Ans:

Low-resolution images contain fewer pixels, making small objects appear blurred or unclear. A low sampling rate reduces spatial details, making it difficult to detect edges and object boundaries accurately. Poor image formation caused by improper focus, lens distortion, or insufficient lighting can further reduce image quality. Increasing the sampling rate, using a higher-resolution camera, and improving image formation conditions help produce clearer images. This improves object visibility and increases the accuracy of object detection systems.